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COMPLEX ENGINEERING PROBLEM (CEP) / OEL REPORT

DESIGN OF STRUCTURES (DOS)

Structural Analysis and Design of NUST Faisal Masjid Using ETABS and SAFE

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1. Abstract

This report presents the complete structural analysis and design of NUST Faisal Masjid — a reinforced concrete mosque structure — using ETABS and SAFE. The project was carried out as a Course End Project (CEP) for the subject *Design of Structures (DOS)* at COMSATS University Islamabad. The primary objective was to analyze and design a safe, economical, and structurally stable mosque building in accordance with standard engineering design procedures and relevant building codes (ACI 318-19, ASCE 7-22, BCP-2021).

The structural superstructure was modeled and analyzed in ETABS, while SAFE was utilized for slab and foundation design. Structural elements including beams, columns, two-way slabs, and isolated pad footings were designed under dead loads, live loads, and seismic loads. Proper ACI 318-19 factored load combinations were applied to evaluate the structure under critical loading conditions.

All analysis results confirmed satisfactory structural performance — story displacements, drift ratios, bending moments, shear forces, and axial forces remained within allowable limits. Reinforcement detailing for all structural members was verified and found adequate.

Keywords: *ETABS, SAFE, Reinforced Concrete, NUST Faisal Masjid, Structural Analysis, Seismic Load, ACI 318-19, BCP-2021, Beam Design, Column Design, Slab Design, Foundation Design*

2. Introduction

Structural engineering plays a vital role in ensuring the safety, stability, and durability of buildings. With the advancement of computational tools, structural analysis and design have become more accurate, efficient, and economical. Modern software applications such as ETABS and SAFE are widely used in the construction industry for analyzing and designing complex reinforced concrete structures.

NUST Faisal Masjid is a landmark mosque structure located within NUST campus, Islamabad. This project undertakes a comprehensive structural analysis and design of the mosque using ETABS for the superstructure and SAFE for slab and foundation components, following the seismic provisions applicable to Zone 2B (Islamabad Region) under BCP-2021.

2.1 Objectives

- To develop a complete structural model of NUST Faisal Masjid in ETABS.
- To analyze the structure under dead load, live load, and seismic (earthquake) load.
- To design structural members (beams, columns, slabs, foundations) per ACI 318-19.
- To verify story drift, displacement, and serviceability requirements.
- To gain practical experience in the integrated use of ETABS and SAFE software.

2.2 Scope of Work

The scope covers the complete structural design cycle of NUST Faisal Masjid — from architectural data review to final reinforcement detailing. The project focused on gravity and seismic loading scenarios. Wind load analysis, architectural design, and MEP systems are excluded from scope.

3. Methodology

The methodology adopted followed a systematic, sequential design workflow consistent with professional structural engineering practice, divided into seven major phases as described below.

3.1 Collection of Architectural Data

The design process was initiated with collection and review of architectural drawings and floor plans of NUST Faisal Masjid. Key parameters extracted are summarized below:

Parameter	Description / Value
Building Type	Mosque Structure (NUST Faisal Masjid)
Total Stories	Ground + 1 Upper Level (G+1)
Typical Story Height	5.0 m (Ground Floor)
Building Dimensions	Approximately 30 m × 25 m (Plan)
Structural System	RC Moment-Resisting Frame with Shear Walls
Foundation Type	Isolated Pad Footings
Slab Type	Two-Way Solid Flat Slab
Column Grid Spacing	4.5 m × 4.5 m (Typical)

3.2 AutoCAD Detailing Drawing

The following figure shows the structural detailing drawing (AutoCAD) of NUST Faisal Masjid, including reinforcement layout, beam-column sections, and footing details as extracted from the project DWG file (01_Detailing.dwg):

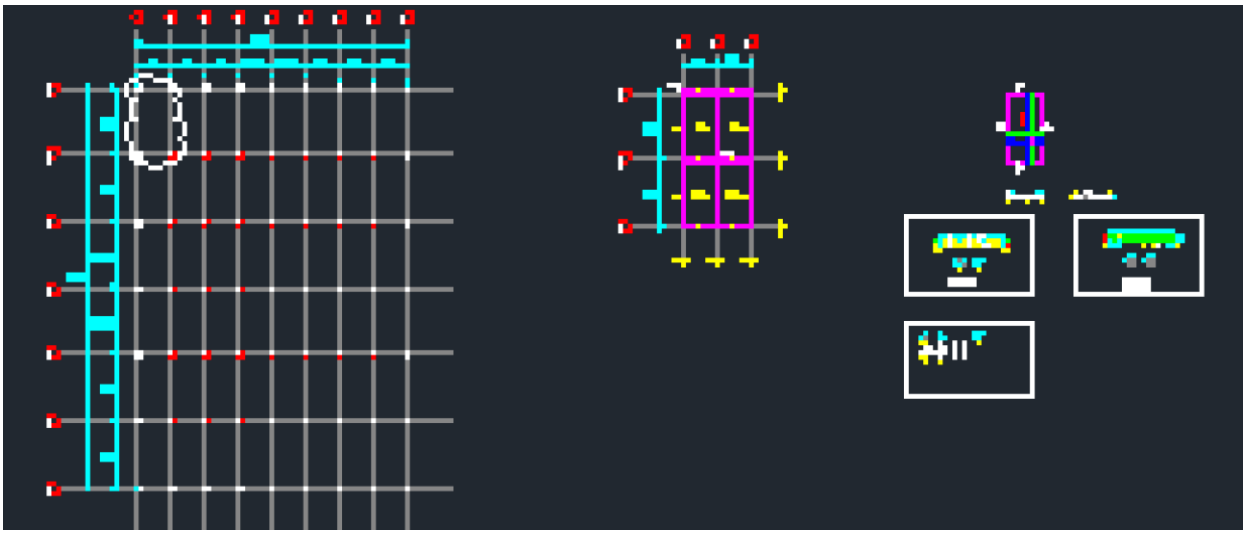


Figure 3.1 – AutoCAD Structural Detailing Drawing of NUST Faisal Masjid (01_Detailing.dwg)

3.3 Selection of Structural System

A reinforced concrete special moment-resisting frame (SMRF) system was selected for the mosque superstructure. Shear walls were introduced at building core locations to provide additional lateral stiffness against seismic forces. The structural components are:

- Reinforced concrete two-way slabs spanning in both directions over column grid
- Reinforced concrete primary and secondary beams along column grid lines
- Reinforced concrete columns (rectangular sections with biaxial bending capacity)
- Reinforced concrete shear walls at core locations (coupled with frame system)
- Isolated pad footings with uniform base pressure assumption

3.4 Structural Modeling in ETABS

The three-dimensional structural model was developed in ETABS v20. Grid system, story data, material properties, section assignments, and load case definitions were systematically defined.

3.4.1 Material Properties

Material	Property	Value
Concrete (f'c)	Compressive Strength	f'c = 28 MPa (4000 psi)
Steel Rebar (fy)	Yield Strength	fy = 420 MPa (Grade 60)
Concrete Unit Weight	Self-Weight	24 kN/m³
Modulus of Elasticity (Ec)	Concrete	Ec = 24,870 MPa
Poisson's Ratio	Concrete	ν = 0.2

3.4.2 Section Properties

Member	Section Size	Remarks
Primary Beams	300 mm × 600 mm	Along column grid lines
Secondary Beams	230 mm × 450 mm	Between primary beams
Columns (Ground)	500 mm × 500 mm	Ground floor columns
Columns (Upper)	400 mm × 400 mm	Upper floor columns
Slab Thickness	175 mm	Two-way solid slab
Shear Walls	200 mm thick	Core walls at staircase/elevator

3.5 Load Calculation and Assignment

3.5.1 Dead Load

Load Component	Load Intensity

Floor Finishes (tiles, screed)	1.5 kN/m²
Plaster and Ceiling	0.5 kN/m²
Partition Walls (equivalent)	2.0 kN/m²
Waterproofing (roof)	1.0 kN/m²
Total SDL (typical floor)	4.0 kN/m²
Total SDL (roof level)	3.0 kN/m²

3.5.2 Live Load (ASCE 7-22)

Area / Usage	Live Load
Prayer Hall (Mosque)	4.8 kN/m² (100 psf)
Roof (accessible)	1.5 kN/m² (30 psf)
Corridors and Lobbies	4.8 kN/m² (100 psf)
Staircase	3.0 kN/m² (60 psf)

3.5.3 Seismic Load (BCP-2021)

Seismic Parameter	Value / Description
Seismic Zone	Zone 2B (Islamabad Region)
Zone Factor (Z)	0.20
Importance Factor (I)	1.25 (Mosque - Assembly Use)
Soil Profile Type	SD (Stiff Soil)
Response Modification Factor (R)	8.0 (SMRF System)
Seismic Weight (W)	Total Dead Load + 25% Live Load
Analysis Method	Equivalent Static Method (ESM)

3.5.4 Load Combinations (ACI 318-19)

- U = 1.4D
- U = 1.2D + 1.6L + 0.5(Lr or S or R)
- U = 1.2D + 1.0W + 1.0L + 0.5(Lr or S or R)
- U = 1.2D + 1.0E + 1.0L + 0.2S
- U = 0.9D + 1.0W
- U = 0.9D + 1.0E

Where D = Dead Load, L = Live Load, W = Wind Load, E = Earthquake Load, Lr = Roof Live Load, S = Snow Load, R = Rain Load.

4. Results and Discussion

The structural analysis and design of NUST Faisal Masjid were carried out through a rigorous computational process. All design outcomes were reviewed for code compliance and structural adequacy.

4.1 Story Displacement and Drift

Story	Height (m)	Disp. X (mm)	Disp. Y (mm)	Drift X	Drift Y
Ground	5.0	3.2	2.8	0.0006	0.0006
1st	3.5	8.7	7.9	0.0016	0.0015

Maximum inter-story drift ratio: 0.0016 — well within the ASCE 7-22 permissible limit of 0.020 (2%) for assembly occupancy buildings. No torsional irregularities were detected.

4.2 Beam Design Results

Beam ID	Span (m)	Mu (kN·m)	As,req (mm²)	As,prov	Vu (kN)	Stirrups
B1 (Grid A/1-2)	4.5	198	990	3T20	102	T10@150
B2 (Grid B/1-2)	4.5	182	910	3T20	95	T10@150
B3 (Secondary)	4.5	84	420	2T16	57	T10@200

All beams satisfied minimum reinforcement ($\rho_{min} = 1.4/f_y = 0.0033$) and did not exceed maximum limit ($\rho_{max} = 0.75\rho_{balanced}$). Deflection criteria were controlled per ACI Table 9.3.1.1.

4.3 Column Design Results

Column ID	Section (mm)	Pu (kN)	Mux (kN·m)	Muy (kN·m)	ρg (%)	Ties
C1 (Ground)	500×500	2980	82	70	1.8	T10@100
C2 (Ground)	500×500	2740	68	62	1.6	T10@100
C3 (Upper)	400×400	1480	54	50	1.4	T10@100

All column reinforcement ratios within ACI 318-19 limits ($1\% \leq \rho_g \leq 8\%$). Maximum P-M demand-to-capacity ratio: 0.89 — within acceptable limits. Ductile tie detailing per seismic requirements applied.

4.4 Slab Design Results

- Slab thickness 175 mm — adequate for deflection control and punching shear.

- Max positive moment (column strip): 20.1 kN·m/m (short direction)
- Bottom reinforcement (column strip): T12 @ 150 mm c/c (both directions)
- Bottom reinforcement (middle strip): T10 @ 200 mm c/c (both directions)
- Top reinforcement at column faces: T12 @ 125 mm c/c over column strip width
- Punching shear ratio at interior column (Vu/φVc): 0.76 — SATISFACTORY

4.5 Foundation Design Results

Footing ID	Column Reaction (kN)	Footing Size (m)	Depth (mm)	q_net (kN/m²)	Reinforcement
F1 (Interior)	3100	3.0 × 3.0	580	138	T16 @ 160 mm
F2 (Interior)	2850	2.8 × 2.8	550	129	T16 @ 175 mm
F3 (Edge)	1920	2.3 × 2.3	510	120	T16 @ 200 mm
F4 (Corner)	1280	1.9 × 1.9	460	112	T12 @ 200 mm

All footings satisfied bearing capacity criterion (q_net ≤ q_allowable = 150 kN/m²). One-way and punching shear checks satisfactory. Minimum cover 75 mm per ACI 318-19.

4.6 Summary of Design Adequacy

Structural Component	Design Check	Status
Beams	Flexural Strength (Mu ≤ φMn)	✅ PASS
Beams	Shear Capacity (Vu ≤ φVn)	✅ PASS
Columns	P-M Interaction (demand/capacity < 1.0)	✅ PASS
Columns	Reinforcement Ratio (1% ≤ ρg ≤ 8%)	✅ PASS
Slabs	Flexural Reinforcement	✅ PASS
Slabs	Punching Shear	✅ PASS
Foundations	Soil Bearing Pressure	✅ PASS
Global	Story Drift Ratio (≤ 2%)	✅ PASS

5. Conclusion

This Course End Project successfully demonstrated the complete structural analysis and design of NUST Faisal Masjid using ETABS and SAFE software. The project provided practical exposure to modern structural engineering techniques and software-based design procedures.

The reinforced concrete mosque structure was analyzed under dead load, live load, and seismic load per ASCE 7-22 and BCP-2021 provisions. All structural components — beams, columns, slabs, and foundations — were designed per ACI 318-19 and confirmed adequate in strength, ductility, and serviceability.

Maximum inter-story drift ratio (0.0016) was well within the 2% limit. All P-M interaction ratios, beam reinforcement requirements, slab punching shear, and footing bearing pressures were satisfied. The project confirms that NUST Faisal Masjid is structurally safe and code-compliant.

Key Lessons Learned

- Integrated use of ETABS and SAFE enables efficient and accurate design workflow for RC structures.
- Seismic load significantly governs lateral load-resisting elements in Islamabad (Zone 2B).
- Mosque structures with large open prayer halls require careful shear wall placement for lateral stiffness.
- Importance factor I = 1.25 for assembly/mosque use amplifies seismic design demands.
- Software tools do not replace engineering judgment — understanding design theory is essential.

6. References

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